MULTI-AGENT ASSISTIVE SYSTEM for Alzheimer's Patients

SENG 696: Agent-Based Software Engineering

Assignment 1
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1. SYSTEM SPECIFICATION

1.1 Project Concept

I propose a hierarchical multi-agent system to assist Alzheimer's patients with daily tasks. The system consists of three specialized worker agents (Audio Agent, Video Agent, and Haptic Agent) coordinated by a higher-level Caregiver Agent. A Reinforcement Learning (RL) validation layer closes the loop by using sensor-derived reward signals to verify that the worker agents are behaving correctly, as shown in Figure 1.

1.2 Business Case

Alzheimer's disease affects over 55 million people worldwide [1]. Patients face progressive cognitive decline that makes daily routines, medication adherence, hydration, safety, and social connection increasingly difficult. Caregivers bear enormous emotional and logistical burdens, often without real-time visibility into the patient's state [2].

Existing assistive apps are single-modality, static, and open-loop: they deliver a reminder but do not verify whether it worked. This project proposes a hierarchical, closed-loop, multi-modal agent system that:

- Delivers every task through audio, video, and haptic channels via three worker agents.
- Observes the patient's behavior through a Caregiver Agent that assigns and supervises tasks.
- Validates agent behavior through a Reinforcement Learning loop driven by camera, audio, and biometric sensors.
- Adapts task delivery in real time: positive patient state yields reward, while distress or physiological instability yields penalty and triggers adjustment.

1.3. Assumption

Caregiver Agent (Robotic Pet)

The Caregiver Agent is assumed to be physically embodied as a therapeutic robotic pet (resembling a dog or cat) that serves as the primary companion and interaction point for the Alzheimer's patient, equipped with cameras embedded in its eyes for facial recognition and scene understanding, microphones for voice and tone detection, haptic sensors and actuators for touch sensing and vibration delivery, biometric sensors for heart rate and motion monitoring, and mobility capabilities to follow, guide, and physically comfort the patient. This robotic pet hosts the Caregiver Agent's observation and task assignment logic onboard, communicates wirelessly with the caregiver panel and cloud analytics, and provides an emotionally engaging, non-threatening interface that reduces patient anxiety while enabling continuous multimodal data collection and task delivery.

Audio Agent

The Audio Agent is assumed to have access to the robotic pet's integrated speaker system and microphone array, enabling it to deliver voice prompts, reminders, calming cues, music therapy, and spoken confirmations through natural-sounding text-to-speech (TTS) synthesis, while also processing vocal tone and ambient audio to support the RL Validation Engine's emotional state classification. It is assumed to support multiple voice profiles (e.g., familiar family voices, caregiver-recorded messages) and can adjust volume, pitch, and speech rate dynamically based on the patient's hearing ability and current cognitive state as determined by RL feedback.

Video Agent

The Video Agent is assumed to utilize the robotic pet's eye-mounted cameras along with any paired tablet or smart display screens in the patient's environment to deliver visual demonstrations, facial recognition of family members and caregivers, scene understanding for safety monitoring (e.g., detecting falls, stove left on), family video messages, and video call functionality. It is assumed to have sufficient onboard or edge-computing GPU resources for real-time facial expression analysis and object detection, with the ability to stream video to the caregiver panel upon request and to render visual content on the robotic pet's body display (if equipped) or nearby screens.

Haptic Agent

The Haptic Agent is assumed to control the robotic pet's vibration motors, tactile feedback mechanisms, and physical movement capabilities to deliver gentle vibrations, success pulses, directional guidance signals (e.g., nudging toward the bathroom or medication), and calming tactile sensations such as a warm pulse or soft purring vibration. It is assumed that the robotic pet's body surface is safe for prolonged human contact, that vibration intensity and patterns are configurable per task and patient sensitivity, and that the Haptic Agent can coordinate with the Caregiver Agent's mobility to provide physical escorting or guiding behavior when the patient needs to move to a different location.

Task Coordinator

The Task Coordinator is assumed to reside as a software module either onboard the robotic pet or on the on-premises edge server, with low-latency access to the RL Validation Engine's policy outputs and the caregiver's task assignments. It is assumed to have a complete task-to-modality matrix mapping all ~20 daily tasks to appropriate combinations of Audio, Video, and Haptic delivery, and that it can dynamically select and sequence modalities based on the patient's current state, time of day, task urgency, and historical success patterns without requiring caregiver intervention for routine decisions.

RL Validation Engine

The RL Validation Engine is assumed to run as a specialized software component either onboard the robotic pet, on the on-premise edge server, or in a hybrid configuration, with access to real-time sensor streams from the robotic pet's cameras, microphones, biometric sensors, and motion detectors. It is assumed to employ a pre-trained reinforcement learning model that classifies patient state into at least three categories (positive/engaged, neutral, distressed/unstable) and computes separate reward signals for emotional state (facial expression, vocal tone, engagement) and physiological stability (heart rate variability, activity level, motion patterns) [3]. It is further assumed that the RL model can be periodically updated through cloud-based federated learning without disrupting real-time operation, and that reward/penalty signals are delivered to the Task Coordinator within a latency threshold suitable for real-time task adaptation (e.g., under 2 seconds).

1.4 Scope

In Scope	**Out of Scope (Wish List)**
Task library of around 20 daily tasks	Medical diagnosis
Three worker agents (Audio, Video, Haptic)	Prescription changes
Caregiver Agent for observation and task assignment	Emergency dispatch integration
RL-based validation using camera, audio, biometric sensors	Multi-patient deployment
Real-time reward and penalty feedback	Custom hardware design

1.5 The Three Worker Agents and Their Modalities

Agent	**Modality**	**What It Does**
Audio Agent	Audio	Voice prompts, reminders, calming cues, music therapy, spoken confirmations
Video Agent	Video	Visual demonstrations, facial recognition, scene understanding, family messages, video calls
Haptic Agent	Haptic	Gentle vibrations, tactile feedback, physical guidance signals, success pulses

1.6 The RL Validation Loop (Core Innovation)

Reward logic:

Positive reward if the patient appears happy (facial expression, vocal tone, engagement) and heart rate or activity is stable.

Negative reward (penalty) if the patient is distressed or physiologically unstable, which forces the worker agents to adjust their delivery.

1.7 Task Library (Representative)

Around 20 common tasks that Alzheimer's patients typically struggle with:

Category	**Example Tasks**
Medication	Morning medication reminder, evening medication reminder, pill count confirmation
Orientation	Time-of-day cue, location cue, name recall prompt
Hygiene	Handwashing guidance, brushing teeth, bathing steps
Meals	Meal assistance, utensil guidance, hydration prompt
Safety	Stove check, door alert, fall detection, emergency call
Wellbeing	Mood check, calming music, memory game, family call
Routine	Wake-up sequence, dressing steps, bedtime preparation

Each task is mapped to one or more of the three worker agents through a task-to-modality matrix.

1.8 Justification for Using MaSE(Multiagent Systems Engineering) Methodology

The MaSE methodology was chosen because its phase-based structure, which includes capturing goals, applying use cases, refining roles, creating agent classes, constructing conversations, assembling internal architectures, and system deployment, maps directly onto the hierarchical, goal-driven, and conversation-centric nature of this assistive system [4]. Its emphasis on goals ensures every agent traces back to a patient-centered objective, its role refinement cleanly separates concerns like audio, video, haptic delivery, coordination, and RL validation, and its formal conversation syntax provides a rigorous vocabulary for specifying the request and response protocols as well as the reward and

feedback protocols central to the closed-loop RL loop. Additionally, MaSE's support for internal agent architecture and deployment modeling allows practical concerns such as onboard processing, edge computing, and federated learning to be addressed within a single unified framework, making the design verifiable, standardized, and extensible for future work.

2. DESIGN DOCUMENT (MaSE)

The MaSE methodology follows seven phases: Capturing Goals, Applying Use Cases, Refining Roles, Creating Agent Classes, Constructing Conversations, Assembling Agent Classes, and System Design. The same phase structure and figure conventions are retained while the diagrams are redesigned for clarity.

MaSE Phase Status

Phase	Output	Status
1. Capturing Goals (Figure 2)	Goal hierarchy with RL validation goal	Complete
2. Applying Use Cases (Figure 3 & 4)	Use case diagram and sequence diagrams	Complete
3. Refining Roles	Role model with worker, caregiver, and RL roles	Complete
4. Creating Agent Classes	Agent class diagram	Complete
5. Constructing Conversations (Figure 5 & 6)	Conversation diagrams for task assignment and RL feedback	Preliminary
6. Assembling Agent Classes (Figure 7)	Agent internal architecture	Preliminary
7. System Design (Figure 9)	Deployment diagram	Future work

2.1 System Architecture (High-Level)

The architecture contains five primary agent classes Caregiver Agent, Task Coordinator, Audio Agent, Video Agent, and Haptic Agent plus the RL Validation Engine as a specialized validation component. These components are organized hierarchically around multimodal task delivery and closed-loop validation.

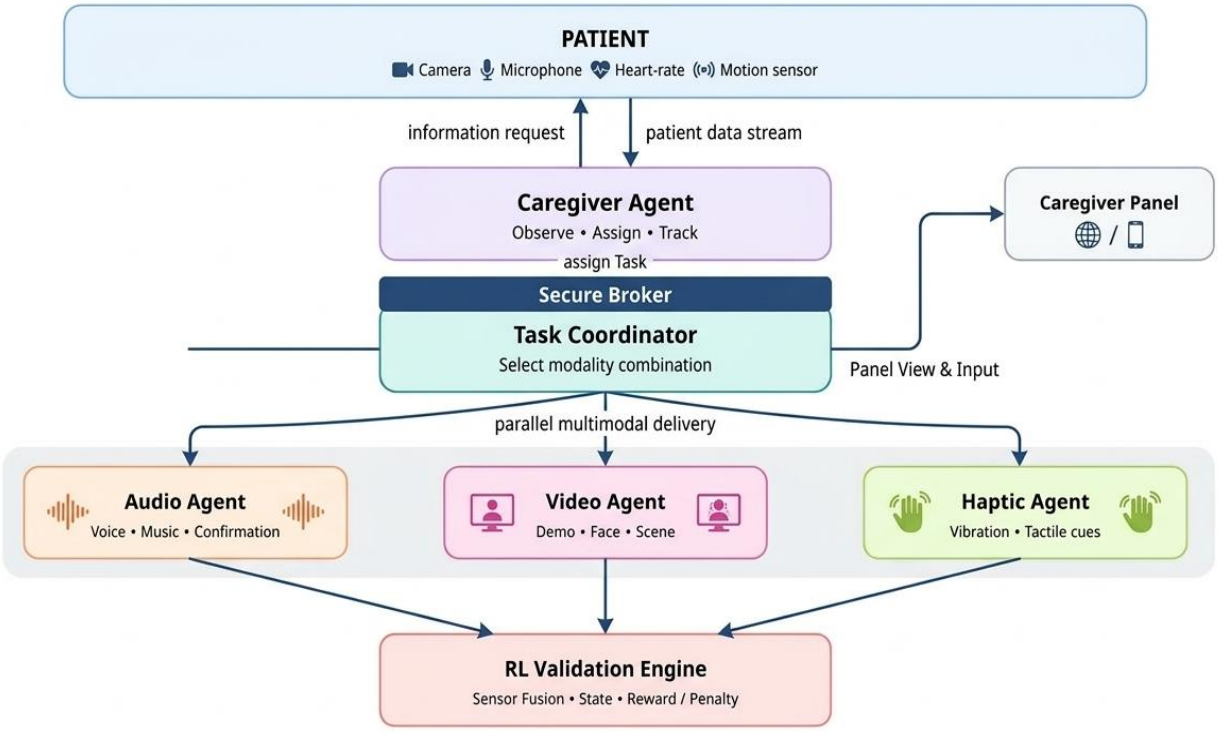


Figure 1. *Hierarchical Multi-Agent Architecture*

2.2 Phase 1: Capturing Goals

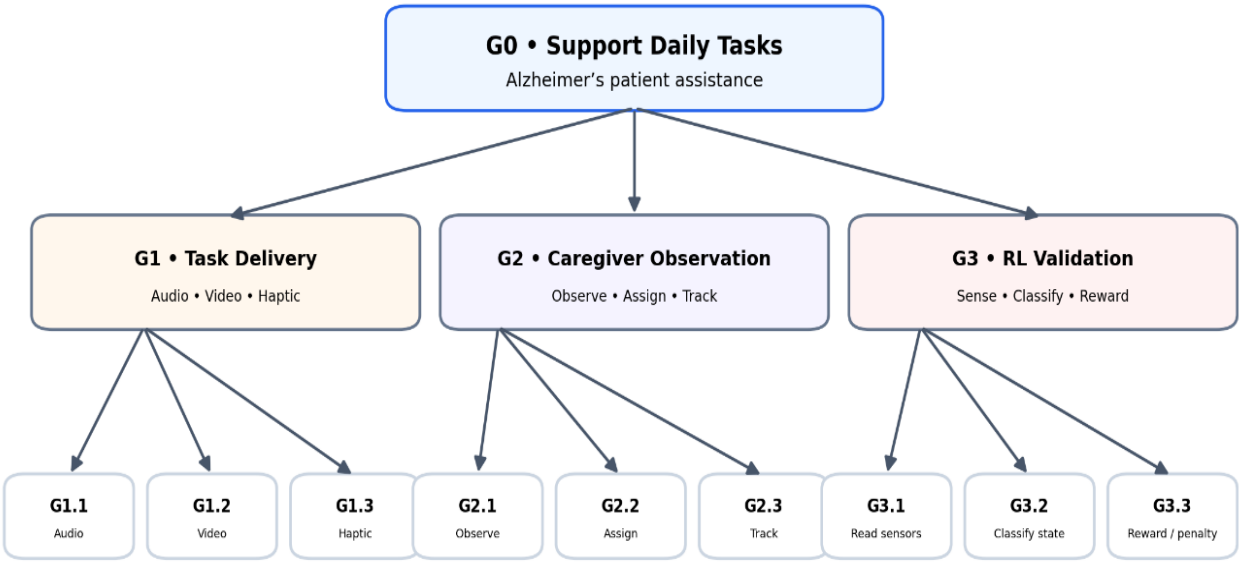


Figure 2. *Goal Hierarchy Diagram*

G1 Task Delivery via Worker Agents: the three worker agents deliver assigned tasks through audio, video, and haptic modalities.

G2 Caregiver Observation: the Caregiver Agent observes patient behavior, assigns tasks from the library, and tracks progress.

G3 RL Validation: the RL Validation Engine reads sensor data, classifies patient state, and issues reward or penalty signals that adjust agent behavior.

Task-to-Modality Matrix

Task	**Audio**	**Video**	**Haptic**
Medication reminder	Voice prompt	Pill animation	Gentle pulse
Handwashing guidance	Step-by-step voice	Demonstration video	Vibration on each step
Orientation cue	Spoken time/place	Visual clock and map	Soft buzz
Meal assistance	Encouragement	Utensil demo	Tap cue
Emergency call	Spoken alert	Flashing SOS	Strong pulse
Calming therapy	Music and voice	Calming visuals	Soft pulse
Memory game	Feedback tone	Game visuals	Tap feedback

2.3 Phase 2: Applying Use Cases

Actors: Patient, Caregiver, Sensors (camera, microphone, heart rate, motion), and RL Validation Engine.

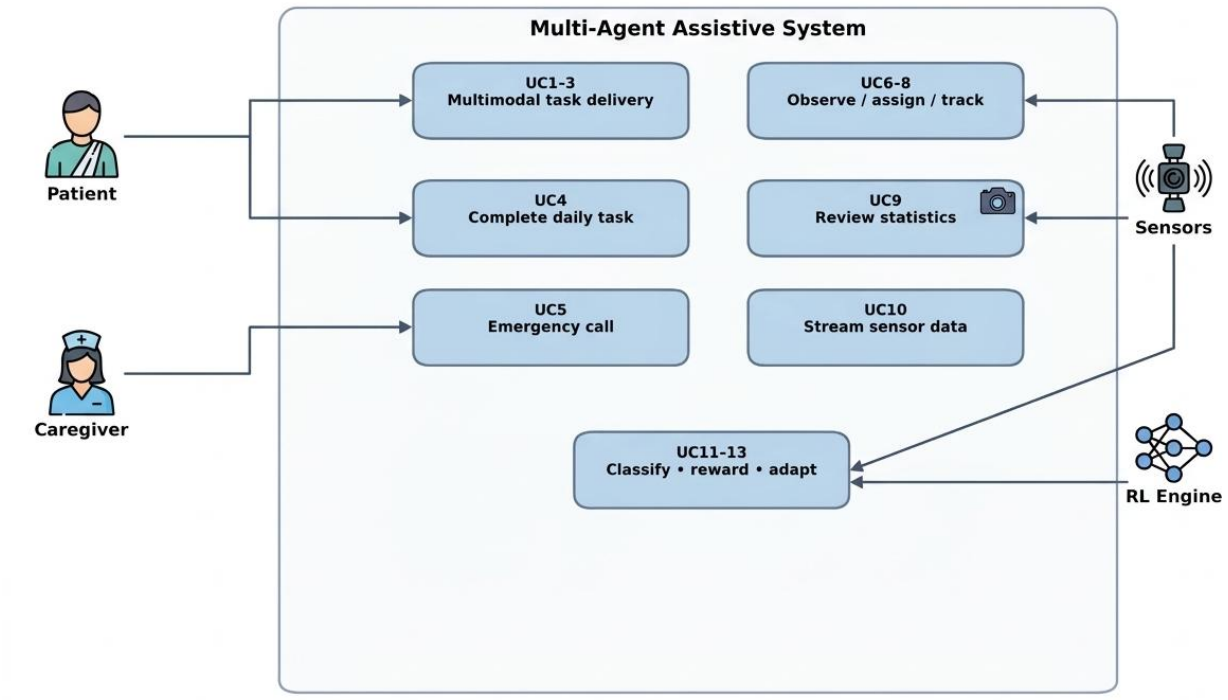


Figure 3. *Use Case Diagram*

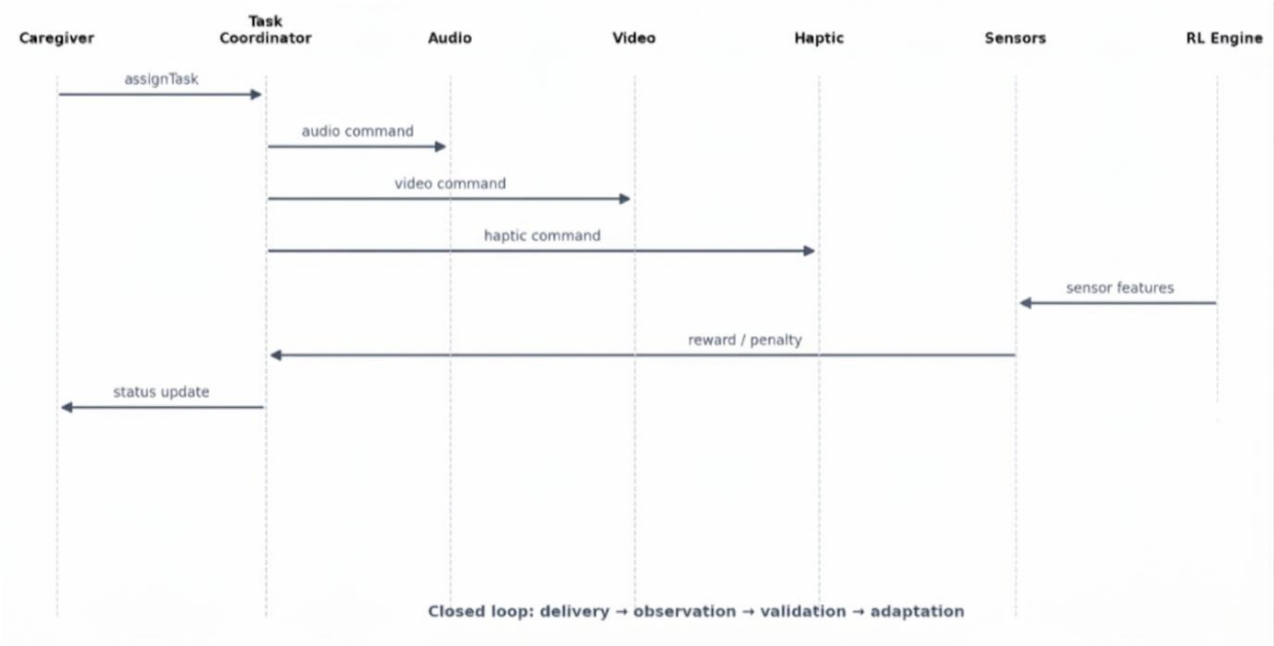


Figure 4. *Sequence Diagram: Task Assignment and RL Validation*

Use Case (UC)	Name	Actor	Modalities	Goal (G)
UC1	Receive task via Audio Agent	Patient	Audio	G1.1
UC2	Receive task via Video Agent	Patient	Video	G1.2
UC3	Receive task via Haptic Agent	Patient	Haptic	G1.3
UC4	Complete daily task	Patient	Audio, Video, Haptic	G1
UC5	Trigger emergency call	Patient	Audio, Haptic	G1
UC6	Observe patient behavior	Caregiver	none	G2.1
UC7	Assign task from library	Caregiver	none	G2.2
UC8	Track task progress	Caregiver	none	G2.3
UC9	Review patient statistics	Caregiver	Video	G2.3
UC10	Stream sensor data	Sensors	none	G3.1
UC11	Classify patient state	RL Engine	none	G3.2
UC12	Issue reward or penalty	RL Engine	none	G3.3
UC13	Adjust agent behavior	RL Engine	none	G3.3

2.4 Phase 3: Refining Roles

Role	Associated Tasks
AudioRole	PlayVoicePrompt(), PlayMusic(), PlayChime(), SpeakConfirmation()
VideoRole	ShowVisualDemo(), RecognizeFace(), UnderstandScene(), PlayVideoMessage()
HapticRole	VibrateGuidance(), PulseOnSuccess(), SoftCalmPulse()
TaskCoordRole	SelectModality(), RouteToWorkers(), ApplyRLPolicy()
CaregiverRole	ObservePatient(), AssignTask(), TrackProgress(), ViewStats()
RL Validation Role	ReadSensors(), ClassifyState(), ComputeReward(), AdjustBehavior()

2.5 Phase 4: Creating Agent Classes

Agent	Responsibility
Caregiver Agent	Observes patient behavior, assigns tasks from the library, tracks progress, and surfaces statistics to the caregiver panel.
Task Coordinator	Receives the assigned task, selects the modality combination based on the RL policy, and routes commands to the three worker agents.
Audio Agent	Plays voice prompts, reminders, calming cues, music, and spoken confirmations.
Video Agent	Displays visual demonstrations, performs facial recognition and scene understanding, and plays family messages and video calls.
Haptic Agent	Drives vibration and tactile feedback, provides physical guidance signals, and generates success pulses.
RL Validation Engine Agent	Reads camera, audio, and biometric sensor streams, classifies patient state, computes reward or penalty, and adjusts agent behavior.

2.6 Phase 5: Constructing Conversations (Preliminary)



***Figure 5**. Conversation: AssignTask*

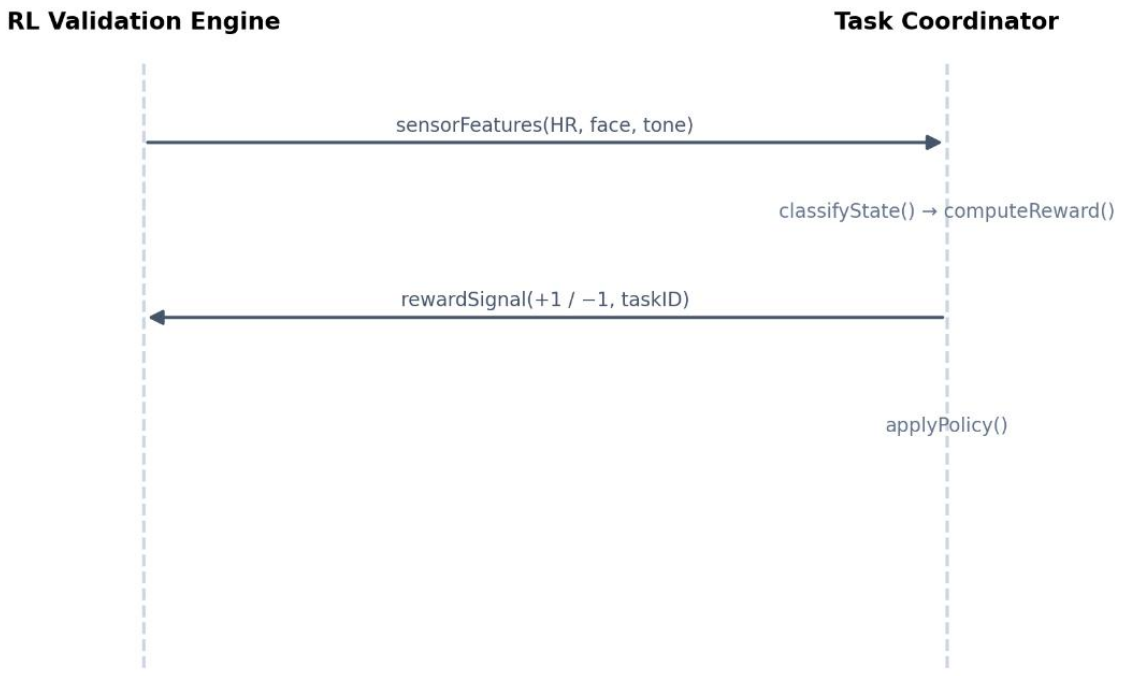


***Figure 6.** Conversation: RewardFeedback*

2.7 Phase 6: Assembling Agent Classes (Internal Architecture)

Figure 7 illustrates the internal processing flow of the Task Coordinator, detailing how incoming inputs are translated into multi-modal agent actions. incoming task assignments and RL reward signals are captured by the Listener and passed to the Interpreter for message parsing. The parsed state is then processed by the Policy Engine, which maps the current RL state to an optimal modality combination. Finally, the Dispatcher routes the resulting execution commands across the central Dispatcher Command Bus directly to the dedicated Audio, Video, and Haptic worker rendering units.

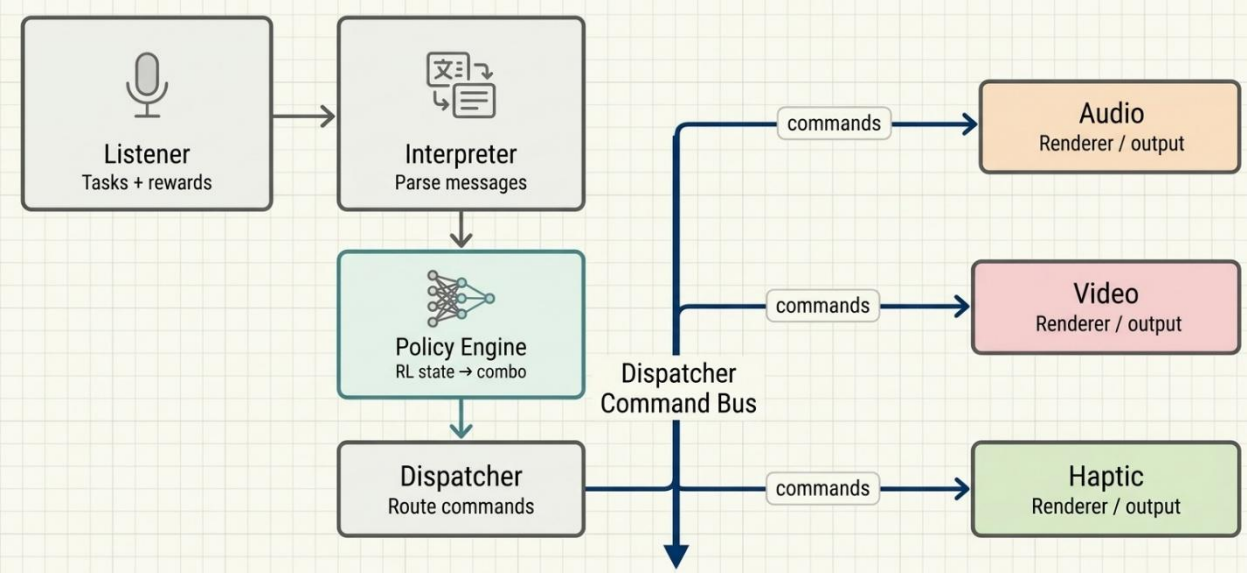


Figure 7. *Task Coordinator Internal Architecture*

Each worker agent follows a similar pattern: Listener, Interpreter, Processor, Renderer.

2.8 Data and Knowledge Sharing Specification

Figure 8 presents the Data and Knowledge Sharing Model, establishing the relational schema required to support real-time task orchestration and continuous reinforcement learning feedback. The Patient profile anchors the dataset, storing demographic and personal attributes, and connects directly to scheduled Task definitions as well as real-time SensorSample records capturing biometric, visual, and vocal features (e.g., heart rate, facial expression, tone, motion). Each task maps to a specific ModalityCombo configuration governing audio, video, and haptic delivery, while task execution generates a TaskOutcome record indicating completion status. Finally, data from the selected modality combination, task outcome, and real-time sensor samples converge into the RewardLog, which records numerical reward signals and policy versions to drive closed-loop model adaptation.

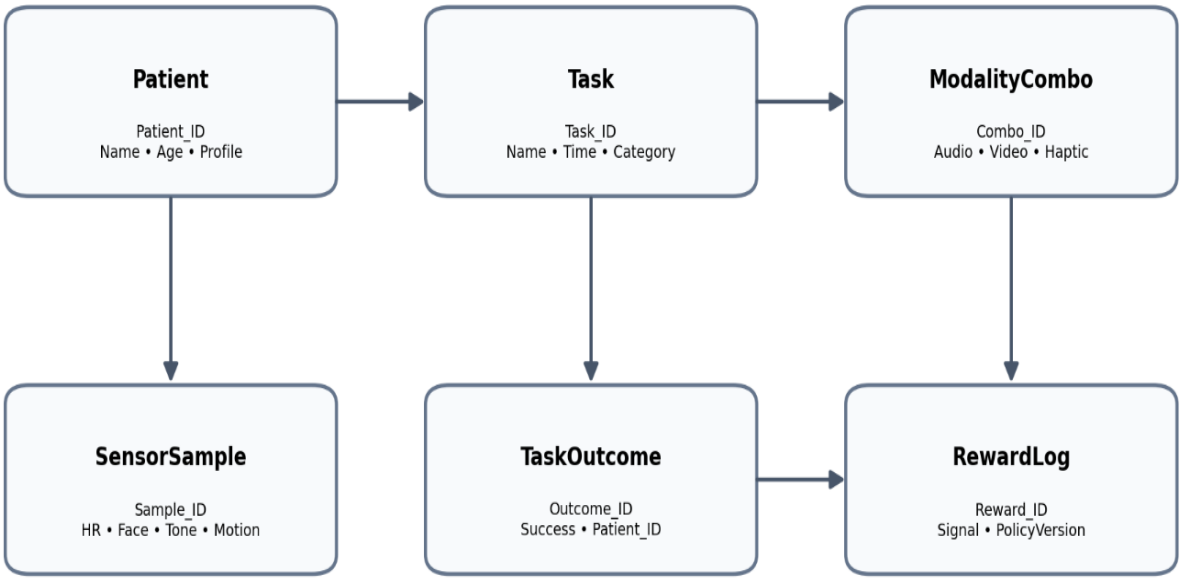


***Figure 8**. Data and Knowledge Sharing Model*

2.9 System Design and Deployment (Future Work)

To transition the multi-agent system into an enterprise-grade, privacy-preserving clinical solution, future work will focus on the deployment infrastructure outlined in Figure 9.

- **Edge Infrastructure and Local Data Fusion:** An on-premises server will serve as a privacy-focused buffer between the patient tablet, embedded sensors, and external networks to perform local data fusion, privacy filtering, and local state monitoring.
- **Cloud Analytics for Policy Training:** A dedicated cloud analytics framework will process long-term reinforcement learning training data transmitted via secure internet links to continually update and refine global policy models over time.
- **Multi-Patient Care Management Dashboard:** The care management dashboard will scale to a multi-patient view, integrating real-time system health monitoring and performance metrics for clinical caregivers across web and mobile panels.
- **Secure Federated Learning:** Decentralized federated learning nodes will securely aggregate local model updates using model cryptography and decentralized hosting, enabling cross-institutional model optimization while maintaining strict patient data privacy.

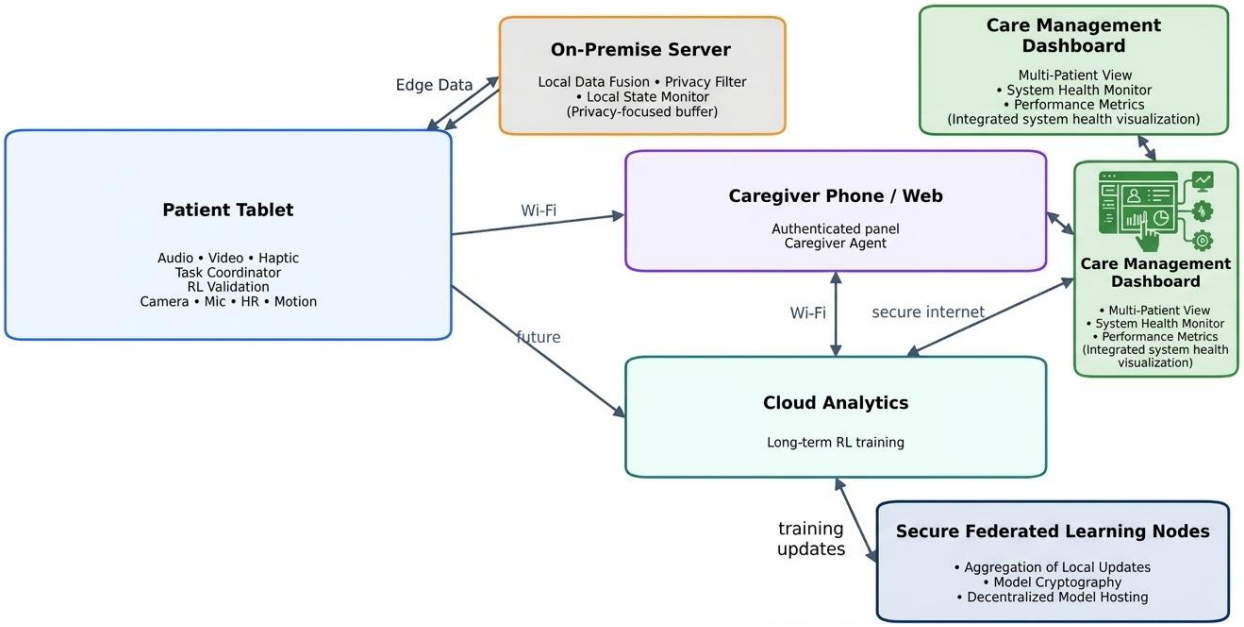


Figure 9. *Deployment Diagram*

3. Why Each Agent Is Necessary

Agent	**Justification**
Caregiver Agent	Top-level observer and task assigner. Selects which of the around 20 tasks the patient needs and dispatches it while keeping the caregiver in the loop.
Task Coordinator	Single decision point for selecting the modality combination for each task based on the RL policy. Avoids duplicating selection logic across workers.
Audio Agent	Encapsulates voice, music, and tone generation. A swappable TTS engine can be used without changing other agents.
Video Agent	Encapsulates rendering, facial recognition, and scene understanding, with an independent lifecycle for GPU and streaming.
Haptic Agent	Encapsulates vibration and tactile feedback and is hardware-specific and swappable across tablet models.
RL Validation Engine Agent	Reads camera, audio, and biometric sensors, classifies patient state, and computes reward or penalty. It remains separate

	because sensor fusion and RL inference have a different compute and latency profile.

4. Design Journal

Input before refinement	Decision	Outcome	Why
Caregiver Agent and three worker agents	Add Task Coordinator and RL Validation Engine	6 agents/components	Without a coordinator, modality selection logic would be duplicated. Without RL validation, the system cannot validate worker behavior.
Generic reward concept	Separate reward for patient emotional state and physiological stability	Two reward dimensions	A patient can complete a task while distressed, and the design aims to penalize that.
Single worker execution	Three parallel worker agents	Multi-modal delivery	Each modality has a different hardware and latency profile.
Static task assignment	Task assignment informed by RL policy	Adaptive	Patient response varies by time, mood, and sensory profile.

5. Summary

The system is a hierarchical multi-agent architecture in which a Caregiver Agent observes the patient and assigns one of around 20 daily tasks to a Task Coordinator, which routes the task to three specialized worker agents (Audio, Video, Haptic). A Reinforcement Learning Validation Engine reads camera, audio, and biometric sensors, classifies the patient's state, and issues reward or penalty signals that adjust worker behavior. Positive patient state and stable heart rate produce reward; distress or instability produces penalty and forces adaptation. This closed-loop RL design makes the system adaptive, empathetic, and verifiable, and it maps onto the MaSE pipeline of goals, roles, agent classes, conversations, internal architecture, and deployment.

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